

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 295124562

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	20.545851 d (ALIASED, see below)	VSX	no match
-> corrected period:	41.091702 d	ASAS-SN	no match
Epoch (t0)	1632.5146 BTJD	Gaia eclipsing-binary table	no match
Duration	0.5000 d	Gaia variable classification	no match
Depth	18.821 % (188,214 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	418.4	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	3 (gate >= 3)		
Robust depth ratio	1.02 (gate >= 0.50)		
Robust depth sigma	257.2 (gate >= 7.0)		
Odd/even z-score	11.67 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase -0.43, 9.8 sigma		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

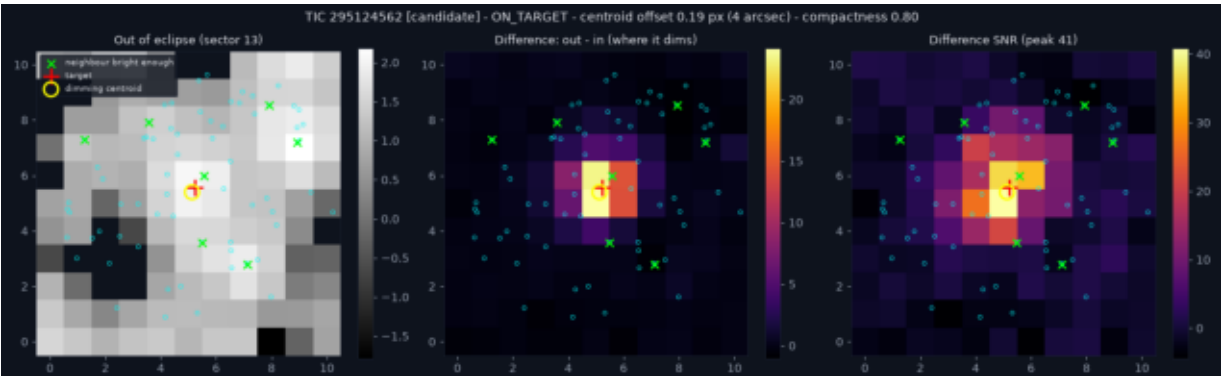
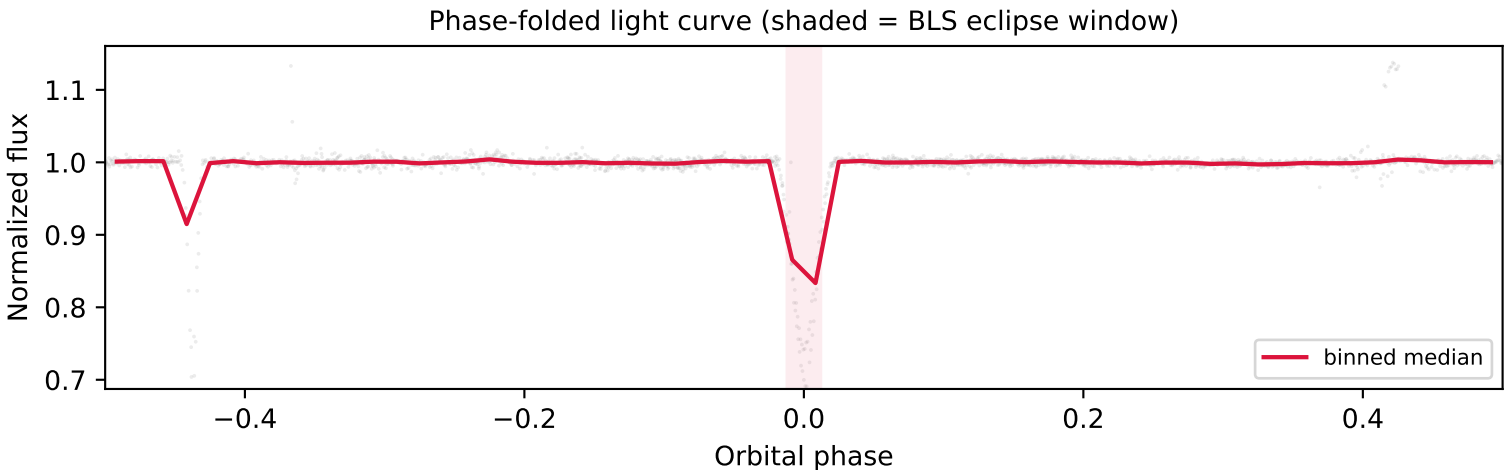
Sector 13 light-curve dip in this sector: 193.6 sigma t0 refined by 0.0000 phase

Difference-image peak SNR 40.86 (gate >= 5.0) Compactness 0.80 (gate >= 0.50)

Dimming centroid offset 0.19 px (4 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 295124561

0.73 px away, dTmag +0.50 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 11.67 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 41.091702 d, in any write-up.

TIC 229832383

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 6.126774 d
Epoch (t0) 1628.8621 BTJD
Duration 0.2440 d
Depth 2.196 % (21,958 ppm)
SNR 97.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 11.2 (gate ≥ 7.0)
Odd/even z-score 0.03 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 12 light-curve dip in this sector: 11.8 sigma t0 refined by 0.0050 phase
Difference-image peak SNR 25.91 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.09 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 229833423
0.54 px away, dTmag +2.83 - farther from the dimming than the target

